

MGMT 405 – Managerial Economics

Teaching Note – Module 3: Hiring Decisions in the Short Run

This note discusses how firms should optimally choose how much of an input to use in the **short run**.

Marginal revenue product (MRP)

MRP is the contribution to revenues from adding one unit of a *factor of production*. For example, consider the standard short-run scenario where capital is fixed while labor (L) is flexible. Then MRP_L tells us how total revenues change if the firm uses more labor. This is the value to the company of one additional unit of labor. The marginal revenue product of labor is defined as:

$$MRP_L = \frac{dTR}{dL}$$

We can rewrite this as (adding dL in the numerator and denominator):

$$\begin{aligned} MRP_L &= \frac{dQ}{dL} \cdot \frac{dTR}{dQ} \\ &= MP_L \cdot MR \end{aligned}$$

Intuitively, this is how much more output the firm gets from one additional unit of labor (MP_L) multiplied by how much revenue the firm gets from one more unit of output (MR).

Marginal labor cost (MC_L) and optimal hiring decisions

The marginal labor cost tells us by how much total costs change when adding one unit of labor. Typically, we think of a unit of labor as a worker. Thus, MC_L is the cost to the company of one additional worker. For the problems we consider in our class, we typically consider firms that are wage takers (i.e., firms that can hire as much labor they would like at the constant market wage). In these cases, MC_L is simply the wage (w). As we saw in class, the optimal choice of labor then takes place where $MRP_L = w$. This means that the firm should add more workers until the worker's contribution to revenue is offset by her/his contribution to costs. The intuition is similar to the $MR = MC$ rule for profit maximization (see the Teaching Note on $MR = MC$). This is the short-run rule: capital is fixed, so labor is the only input being adjusted. The long-run counterpart, where capital can be adjusted too, is in the Teaching Note on the Bang-for-the-Buck Rule.

If $MRP_L < w$, adding an additional worker is a bad decision because revenues grow less than (labor) costs, thus lowering profits. If $MRP_L > w$, adding an additional worker increases total profit.

In some instances, the firm's choice of labor may not be continuous. For example, the firm may be deciding between 5 and 6 workers (a discrete choice), instead of 5 and 5.8 workers. In this case, the firm should hire the 6th worker if $MRP_L \geq MC_L$.

We also briefly discussed the case of wage searchers, who may have to raise their wage offers in order to attract more workers (for example, highly specialized workers such as surgeons or top designers). In these cases, the marginal labor cost MC_L is not constant. An additional worker should be hired if $MRP_L \geq MC_L$.

The reason is that hiring one more worker bids up the wage for everyone already employed, so the full marginal cost of that worker is larger than the wage she or he is paid. In the example we did in class, the third star designer is paid \$2M, but the two existing designers each get a \$0.5M raise, so the marginal labor cost of the hire is \$3M rather than \$2M.